

Prodip Sengupta

AI-Powered Full-Stack Developer · MERN · LLM Agents · RAG Pipelines · Generative AI
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PROFILE

Full-stack developer specialising in MERN and Generative AI, with hands-on experience shipping multi-agent systems, RAG pipelines, and tool-calling LLM applications using Claude, GPT-4o, and Gemini. Delivered end-to-end AI products covering LLM orchestration, vector databases, streaming UIs, and cloud deployment. Targeting Fullstack AI Dev, Jr. Gen AI Engineer, and FDE roles.

EXPERIENCE

Frontend Developer & Project Manager

Nov 2025 – Apr 2026

Tara Effects Photography · Freelance · tara-effects.vercel.app

- Delivered a production Next.js site (photography, videography, digital invitations) with Framer Motion transitions and WhatsApp Business API booking — **Lighthouse 95+ on mobile**, avg image load <1s via Cloudinary.
- Owned full delivery lifecycle: requirements gathering, design sign-off, and Vercel deployment as sole developer and project manager.

PROJECTS

VitalTrace — Agentic AI Health Intelligence Platform

Jun 2026

Next.js 14 · LangChain · Supabase · pgvector · Gemini / Groq / NVIDIA NIM · Framer Motion

[Live Demo](#) | [GitHub](#)

- Engineered a **3-agent LangChain pipeline** (Extractor → Analyzer → Pattern Detector) across **Gemini, Groq, and NVIDIA NIM** that parses diagnostic PDFs from Dr Lal PathLabs, Thyrocare, and Metropolis — structuring biomarkers, flagging anomalies, and storing embeddings via **pgvector**.
- Implemented **RAG-powered longitudinal trend detection** over a Supabase vector store, auto-generating physician consult briefs and personalised lifestyle recommendations grounded in the patient's historical records.

FinDoc AI — RAG-Powered SEC Filing Research Assistant

Jun 2026

React · Node.js · Express · LangGraph JS · Supabase · pgvector · OpenAI

[Live Demo](#) | [GitHub](#)

- Engineered a **hybrid retrieval pipeline** fusing pgvector semantic search and Postgres full-text search via **Reciprocal Rank Fusion** — cutting analyst document-intake **from hours to seconds** with a re-ranking pass.
- Orchestrated a **LangGraph ReAct agent** with iterative tool-calling, citation log validation (hallucination guard), and structured JSON output with numbered markers tied to exact document UUIDs.

GAIA Benchmark Agent — Stateful Multi-Step LLM Reasoner

May 2026

LangGraph JS · Node.js · Gemini 2.5 Flash · GPT-4o (Azure) · Qwen-2.5 · Docker

[Hugging Face Space](#) | [GitHub](#)

- Built a stateful cyclic LangGraph agent with a tool layer integrating sandboxed Python execution, web grounding, Wayback Machine, and multimodal analysis (text, table, audio, image).
- Implemented a **resilient model fallback pool** (Gemini → GPT-4o → Qwen); achieved **35% on GAIA Level-1** (SOTA ≈40%) with a Dockerized HF Spaces deployment.

LeafCart — AI-Powered Grocery Delivery Platform

Mar 2026

MERN · Gemini Function Calling · Stripe · JWT

[Live Demo](#) | [GitHub](#)

- Built a full-stack MERN platform (50+ products, dual seller/admin dashboards, real-time inventory) with a **Gemini function-calling AI cart agent** that interprets natural-language requests and queries MongoDB autonomously.
- Secured with JWT auth and Stripe payments; deployed on Vercel with **avg ~600 ms API response** and a horizontally scalable MongoDB backend.

SKILLS

Languages: JavaScript (ES2024), TypeScript, Python, Java, HTML5, CSS3

Frameworks: React.js, Next.js, Node.js, Express.js, Tailwind CSS, Redux, Framer Motion

Generative AI: OpenAI, Gemini, Claude, LangChain, LangGraph, RAG Pipelines, AI Agents, Tool Calling, Prompt Engineering, Embeddings, Vector Search

Databases: MongoDB, PostgreSQL, Supabase, pgvector, Pinecone

Testing & Eval: Jest, React Testing Library, Postman, LangSmith, RAGAS

Tools & Infra: Git, GitHub, Docker, Vercel, Render, Cloudinary, Stripe API

EDUCATION

Master of Computer Applications

Aug 2025

Maulana Abul Kalam Azad University of Technology · CGPA 7.80 / 10

Bachelor of Computer Applications

Jul 2023

University of North Bengal · **CGPA 9.07 / 10**

CERTIFICATIONS

Claude Code in Action — Anthropic ([Verified](#))

Mar 2026

AI Agents Course (Certificate of Excellence) — Hugging Face ([Verified](#))

Claude 101 — Anthropic ([Verified](#))

Mar 2026

Introduction to Model Context Protocol — Anthropic ([Verified](#))

May 2026

Jun 2026